

Table 3: Description of the mandatory Event CDF fields. Every recorded event must include all these fields. The coordinate system is defined in Paragraph 3.3 where the x axis is the sidelines and the y axis the goal lines. Note that {i} is used as an index for the items in a list.

Root	Field	Description	Type
match	id	Unique match identifier	String
meta	is_synced	Indicates if synced tracking data is available (true) or not (false) for this event	Boolean
event	id	Unique identifier of the event	String
event	time	Absolute time in UTC of when the event started (e.g., moment the pass is given, the moment the ball leaves the hands on a thrown in)	String
event	period	Period of the match which can be first_half, second_half, first_half_extretime, second_half_extretime, or shootout	String
event	type	Name of the event type (e.g. shot, pass, referee, defending, goalkeeping, misc etc.)	String
event	sub_type	Name of the event sub type, which can be for shot - (penalty_kick, free_kick, corner_kick etc.); pass - (throw_in, free_kick, corner_kick, goal_kick, kick_off etc.); referee - (final_whistle, foul, caution, offside, substitution, player_on, player_off. Player on and player off events occur for example when a player has to abandon the game due to a lack of available substitutions, or as a temporary measure after a medical check); defending - (clearance, interception, pressing, block etc.); goalkeeping - (save, save_attempt, claim, etc.); misc - (other_ball_action, chance_without_shot, tackle, etc.)	String / null
event	is_successful	Denotes whether the event was successful (true) or not (false)	Boolean
event	outcome_type	Event outcome options: shot - (successful, saved, blocked, wide, woodwork, own_goal); pass - (e.g. successful, out_of_play, intercepted); referee - (start, end, injury, offside, yellow_card, red_card or second_yellow_card); defending - (e.g. clearance, block, interception, pressure); goalkeeping (e.g. save, save_attempt, claim); misc - (e.g. successful, unsuccessful)	String
event	player_id	Unique identifier of the player performing the action. For example, player committing a pass, or making a tackle. Required for all non-referee events. Referee events require event/official_id.	String
event	team_id	Unique team identifier of the player performing the action. Required for all non-referee events. Referee events require event/official_id.	String
event	receiver_id	Unique identifier of the player receiving a pass. Leave <i>null</i> when the event is not a pass or when the pass has no receiver.	String / null
event	receiver_time	Absolute time in UTC of the first moment the ball was received. Leave <i>null</i> when the event is not a pass or when the pass has no receiver.	String / null
event	x	x location where the action of player_id happened (m).	Float
event	y	y location where the action of player_id happened (m).	Float
event	x_end	x location where the action of player_id ended (m), null for single-point events that have no end coordinate.	Float / null
event	y_end	y location where the action of player_id ended (m), null for single-point events that have no end coordinate.	Float / null
event	body_part	Denotes the body part used by player_id, null allowed for events other than pass and shot.	String / null